

Part VII — Fluids · Chapter 35

Density and Pressure

How mass per volume and force per area govern all fluid behavior

Press your finger into water: you feel no resistance, yet the ocean floor is under crushing pressure that would destroy a submarine not built for it. A steel ship floats while a solid steel ball sinks to the bottom of a pond. A hydraulic car lift raises a vehicle with a fraction of the force it would take to lift it directly. Divers experience increasing pressure with each metre of descent; astronauts experience zero atmospheric pressure in orbit. All of these phenomena are explained by two quantities introduced in this chapter: density — how much mass is packed into a given volume — and pressure — how much force is exerted on each unit of area. These are the foundational concepts that make all of fluid mechanics comprehensible.

35.1 What a Fluid Is

A fluid is any substance that can flow and deform continuously under the application of a shear stress (a force applied parallel to a surface). Both liquids and gases are fluids. Solids, by contrast, resist shear deformation — they can be pushed without flowing.

The defining property of a fluid is the inability to sustain a static shear stress: apply a sideways force to a layer of fluid and it will flow, no matter how small the force. This distinguishes water from ice (a solid that resists shear), and air from glass (which behaves like an extremely viscous fluid over geological timescales, despite seeming solid on human timescales).

Fluids do resist compression and tension (normal stresses), which is why you cannot push your hand through concrete even though concrete technically flows over centuries. But for all practical purposes of this chapter and the next two, fluids are treated as substances that transmit pressure forces in all directions and deform without limit when sheared.

35.2 Liquids vs. Gases

Liquids and gases share the property of being fluids, but they differ in important ways

that arise from their microscopic structure.

A liquid has a definite volume but no definite shape. Its molecules are close together and interact through intermolecular forces, but they can slide past each other (unlike a solid). Liquids are nearly incompressible: changing the pressure by a factor of two changes the volume by much less than 1%. This makes liquids excellent for hydraulic systems, where pressure must be transmitted with minimal energy storage in compression.

A gas has neither definite volume nor definite shape. Its molecules are widely separated and interact very weakly. Gases are highly compressible — halving the volume roughly doubles the pressure (Boyle's Law, Chapter 32). Gases fill any container entirely. These properties make gas behaviour strongly dependent on pressure and temperature in ways that liquid behaviour is not.

Property	Liquid	Gas
Definite volume?	Yes	No — fills container
Definite shape?	No — takes container shape	No — fills container
Compressible?	Negligibly (treat as incompressible)	Yes, significantly
Density	High (hundreds to thousands kg/m^3)	Low (fraction to few kg/m^3)
Molecular spacing	Close together	Far apart
Free surface?	Yes (liquid–gas interface)	No free surface

Table 35.1 Comparison of liquid and gas properties relevant to fluid mechanics.

35.3 Density

Density (ρ , rho) is the mass per unit volume of a substance. It is an intensive property — it does not depend on how much material is present, only on what the material is and its thermodynamic state (temperature, pressure).

$$\rho = m / V$$

Density ρ (kg/m^3) = mass m (kg) $\div$ volume V (m^3). The SI unit is kg/m^3 . Common non-SI unit: g/cm^3 ($1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$).

Density measures how tightly mass is packed. Dense materials (lead, steel, gold) have large amounts of mass in small volumes. Less dense materials (wood, foam, air) have small amounts of mass in larger volumes. Water has a density of almost exactly 1000 kg/m^3 (or 1 g/cm^3) at 4°C — this is not a coincidence but was built into the original definition of the kilogram.

Substance	$\rho \text{ (kg/m}^3\text{)}$	$\rho \text{ (g/cm}^3\text{)}$	Notes
Osmium (densest element)	22 590	22.59	Densest naturally occurring element
Gold	19 300	19.3	Heavy precious metal
Lead	11 340	11.34	Dense industrial metal
Iron / steel	7 870	7.87	Common structural metal
Aluminium	2 700	2.70	Lightweight structural metal
Glass	2 500	2.50	Approximate; varies with type
Seawater	1 025	1.025	Slightly denser than fresh water
Water (4°C)	1 000	1.000	Reference density for specific gravity
Ice (0°C)	917	0.917	Less dense than liquid water
Ethanol	789	0.789	Common organic solvent
Air (20°C , 1 atm)	1.204	0.001204	Much less dense than liquids
Helium (20°C , 1 atm)	0.164	0.000164	Lightest noble gas

Table 35.2 *Densities of common substances at approximately room temperature and atmospheric pressure.*

35.4 Calculating Density

Density calculations involve three possible unknowns: density, mass, and volume. Given any two, the third is determined by $\rho = m/V$, $m = \rho V$, or $V = m/\rho$.

Step 1: Identify: What is known (two of ρ , m , V) and what is asked (the third).

Step 2: Units: If density is in kg/m^3 , mass must be in kg and volume in m^3 . If density is in g/cm^3 , mass in g and volume in cm^3 .

Step 3: Compute: Apply the formula. Note: $1 \text{ mL} = 1 \text{ cm}^3$; $1 \text{ L} = 1000 \text{ cm}^3 = 10^{-3} \text{ m}^3$.

Example — Mass from density A solid gold bar measures $25 \text{ cm} \times 8 \text{ cm} \times 4 \text{ cm}$. Volume = $800 \text{ cm}^3 = 800 \times 10^{-6} \text{ m}^3$. Mass = $\rho V = 19\,300 \times 800 \times 10^{-6} = 15.44 \text{ kg} \approx 15.4 \text{ kg}$.

Example — Density from mass and volume A 250 mL bottle contains 198 g of ethanol. Volume = 250 cm^3 . $\rho = 198/250 = 0.792 \text{ g/cm}^3 = 792 \text{ kg/m}^3$. (Table value: 789 kg/m^3 — close, consistent with real ethanol.)

For irregular objects, volume can be found by water displacement: submerge the object in a graduated cylinder and measure the volume of water displaced. This technique was famously used by Archimedes to check whether a crown was pure gold.

35.5 Relative Density and Specific Gravity

Specific gravity (SG), also called relative density, is the ratio of the density of a substance to the density of a reference substance, usually water at 4°C :

$$\text{SG} = \rho_{\text{substance}} / \rho_{\text{water}} = \rho_{\text{substance}} / 1000$$

(where ρ is in kg/m^3)

Specific gravity is dimensionless. $\text{SG} < 1$ means the substance floats on water; $\text{SG} > 1$ means it sinks. Water itself has $\text{SG} = 1.000$.

Specific gravity is convenient because its numerical value in the metric system equals the density in g/cm^3 and makes comparisons immediate: $\text{SG} = 0.92$ means the substance is 92% as dense as water (ice floats); $\text{SG} = 7.87$ means iron is 7.87 times denser than water (iron sinks). No unit conversion is needed.

Specific gravity is measured by a hydrometer: a weighted float that sinks deeper in denser liquids. Hydrometers are used to check battery acid concentration, the sugar content of wine musts (brix measurement), antifreeze concentration, and the salinity of aquaria.

35.6 Pressure

Pressure is the force exerted perpendicular to a surface per unit area of that surface. It is one of the most important quantities in fluid mechanics and connects the macroscopic behaviour of fluids directly to the forces they exert.

$$P = F / A$$

Pressure P (Pa) = perpendicular force F (N) $\div$ area A (m^2). $1 \text{ Pa} = 1 \text{ N/m}^2$. Pressure is a scalar: it acts equally in all directions at a point in a fluid.

The same force applied over a smaller area produces greater pressure. This is why a stiletto heel sinks into soft ground while a flat boot does not: the same body weight (force) is concentrated into a much smaller area by the heel. Why a sharp knife cuts more easily than a dull one: the same force acts over a much smaller edge area, producing very high pressure. Why snowshoes allow walking on snow: the body weight is spread over a large area, keeping the pressure below the snow's yield strength.

Example — Pressure calculation A 60 kg woman stands on one stiletto heel (contact area $\approx 0.5 \text{ cm}^2 = 0.5 \times 10^{-4} \text{ m}^2$). $P = mg/A = (60 \times 9.81)/(0.5 \times 10^{-4}) = 588/0.00005 = 11.8 \text{ MPa}$. This is comparable to the yield strength of some soft materials — hence the damage stiletto heels cause to wooden floors.

35.8 Fluid Pressure and Depth

In a static fluid (one not in motion), pressure increases with depth because the weight of the fluid above presses down on the fluid below. The pressure at any depth h below the free surface of a fluid of density ρ is:

$$P = P_0 + \rho gh$$

Hydrostatic pressure equation. P = absolute pressure at depth h ; P_0 = pressure at the surface (typically atmospheric pressure); ρ = fluid density (kg/m^3); $g = 9.81 \text{ m/s}^2$; h = depth below surface (m).

This equation has several key consequences. First, pressure depends only on depth, not on the shape of the container or the horizontal position. A point 5 m below the surface of a narrow well and 5 m below the surface of the ocean both have the same pressure (if the same fluid and same surface pressure). This is a non-obvious result that puzzled natural philosophers until Pascal systematised it.

Second, pressure increases linearly with depth. Every additional metre of water adds $\rho gh = 1000 \times 9.81 \times 1 \approx 9810 \text{ Pa} \approx 0.097 \text{ atm} \approx 1 \text{ psi}$ of pressure. At 10 m depth, water pressure is approximately 1 atm, so the total absolute pressure is about 2 atm (atmospheric plus water). At 100 m, about 11 atm. At the deepest point of the ocean (Challenger Deep, $\approx 11 \text{ km}$), the pressure is about 1100 atm.

Example — Pressure at depth Find the absolute pressure at the bottom of a 4.0 m deep swimming pool. $P_0 = 101\,325 \text{ Pa}$. $\rho = 1000 \text{ kg/m}^3$. $P = 101\,325 + 1000 \times 9.81 \times 4.0 = 101\,325 + 39\,240 = 140\,565 \text{ Pa} \approx 140.6 \text{ kPa} \approx 1.39 \text{ atm}$.

Key Point: Pressure at a given depth is the same in all directions — up, down, and sideways. This follows from Pascal's principle: a fluid cannot sustain a shear stress, so the pressure at any point must be isotropic (the same in all directions). A submerged object therefore experiences pressure pushing on it from all sides.

35.9 Atmospheric Pressure

Atmospheric pressure is the pressure exerted by the weight of the column of air above a given point in the atmosphere. At sea level, standard atmospheric pressure $P_{\text{atm}} = 101\,325\text{ Pa} = 101.325\text{ kPa} \approx 1\text{ atm} \approx 760\text{ mmHg} \approx 14.7\text{ psi}$.

Atmospheric pressure decreases with altitude because the column of air above is shorter and less dense. At 5500 m altitude (approximately the height of the summit of Kilimanjaro), atmospheric pressure is roughly half of sea-level value — each breath contains half as many oxygen molecules as at sea level. At 10 000 m (typical cruising altitude for aircraft), pressure is about 26 kPa, which is why aircraft cabins are pressurised.

Atmospheric pressure varies with weather: high-pressure systems (anticyclones) have slightly above-average P_{atm} ; low-pressure systems (cyclones, storms) have below-average P_{atm} . A falling barometric pressure reading signals approaching unsettled weather. The variation is typically 1–2% around the standard value, significant for aviation and precision measurement.

The weight of the entire atmosphere acts on every surface. A horizontal flat surface of area 1 m^2 has about $101\,325\text{ N} \approx 10$ tonnes of air weight pressing down on it at sea level. This force is balanced by the atmospheric pressure acting upward on the underside of any real surface, so we do not feel it as a net force. When pressure is removed from one side (as in a vacuum seal or suction cup), the net force becomes apparent.

35.10 Gauge Pressure vs. Absolute Pressure

Absolute pressure is pressure measured from zero (a perfect vacuum). It is always non-negative.

Gauge pressure is pressure measured relative to atmospheric pressure: $P_{\text{gauge}} = P_{\text{abs}} - P_{\text{atm}}$. Gauge pressure can be positive (pressure above atmospheric, as in an inflated tyre) or negative (pressure below atmospheric, as in a partial vacuum or a suction cup). Gauge pressure is zero when the absolute pressure equals atmospheric.

$$P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}$$

Relationship between absolute and gauge pressure. $P_{\text{atm}} \approx 101.325 \text{ kPa} = 1 \text{ atm} = 14.696 \text{ psi}$.

Most everyday pressure gauges read gauge pressure: a flat tyre reads 0 psi gauge, not 14.7 psi absolute. A tyre inflated to "32 psi" has gauge pressure 32 psi and absolute pressure $32 + 14.7 = 46.7 \text{ psi}$. Always clarify which type of pressure is being used; using gauge where absolute is required (or vice versa) causes errors in gas law and fluid pressure calculations.

Study Tip: Gas law calculations (Chapters 32–34) require absolute pressure. Fluid depth-pressure calculations using $P = P_0 + \rho gh$ require absolute pressure at the surface ($P_0 = P_{\text{atm}}$ for an open container). Most real-world gauges read gauge pressure. Always convert before applying formulas.

35.11 Pascal's Principle

Pascal's Principle states: a pressure change applied to a confined fluid is transmitted undiminished to every part of the fluid and to the walls of the container.

This follows directly from the hydrostatic pressure equation $P = P_0 + \rho gh$. If P_0 changes (for example, by applying a force to a piston), then P changes by the same amount at every point in the fluid (since ρgh is unchanged for any fixed depth h). The pressure change propagates instantaneously throughout the entire fluid volume.

Pascal's principle makes fluids ideal for transmitting forces over distances and around corners without mechanical linkages. It is the operating principle behind every hydraulic system: brakes, lifts, presses, and jacks.

Example — Pascal's principle A pump increases the pressure at one end of a hydraulic line by 500 kPa. The pressure at a cylinder 10 m away (same elevation) increases by exactly 500 kPa. The 10 m distance does not reduce the transmitted pressure change.

35.12 Hydraulic Systems

A hydraulic system transmits force from one piston to another through an incompressible fluid. The key insight is that pressure is the same throughout the fluid (Pascal's principle), but the force exerted by the fluid depends on the area of the piston. A small force applied to a small piston can be amplified into a much larger force at a large piston.

$$P = F_1/A_1 = F_2/A_2 \quad \Rightarrow \quad F_2 = F_1 \times A_2/A_1$$

Hydraulic force amplification. The ratio A_2/A_1 is the hydraulic mechanical advantage. A large ratio allows a small force to lift a large load.

The mechanical advantage is A_2/A_1 . If the output piston is 10 times larger in area than the input piston, the output force is 10 times the input force. But energy is conserved: the output piston moves a shorter distance. Specifically, $A_1 d_1 = A_2 d_2$ (fluid volume is conserved), so $d_2 = d_1 A_1/A_2$. Force is amplified by A_2/A_1 ; distance is reduced by the same factor. Work done in = work done out (in the ideal case).

Example — Hydraulic lift A car lift has a small input piston (area 5.0 cm^2) and a large output piston (area 200 cm^2). A mechanic pushes the input piston with 150 N . Output force: $F_2 = 150 \times 200/5.0 = 6000 \text{ N}$. This is enough to lift a car of mass $6000/9.81 \approx 612 \text{ kg}$. The mechanical advantage is $200/5 = 40$.

Hydraulic systems power car brakes (master cylinder amplifies foot force to disc/drum pressure), aircraft control surfaces (pilots move controls against minimal resistance; hydraulics drive the actual surface), construction machinery (excavators, bulldozers), and industrial presses (forging, stamping, injection moulding). The combination of Pascal's principle and incompressibility makes hydraulic force transmission essentially lossless at practical engineering speeds.

35.13 Common Mistakes in Density and Pressure Problems

- Confusing mass and weight. Density = mass/volume, not weight/volume. The weight of a fluid does appear in the pressure formula $P = \rho gh$ (as ρg = weight per unit volume), but density itself is a mass property.
- Using gauge pressure when absolute pressure is required. The hydrostatic formula $P = P_0 + \rho gh$ gives absolute pressure. If P_0 is gauge (zero for open containers), you must add P_{atm} to get absolute. Failing to include P_{atm} leads to underestimating pressure by about 101 kPa .
- Forgetting to convert units consistently. If density is given in g/cm^3 and depth in metres, convert before applying ρgh . Either convert ρ to kg/m^3 and use g in m/s^2 , or convert h to cm and use g in cm/s^2 (inconvenient). Easiest: always use SI units (kg , m , Pa).
- Thinking pressure depends on the shape of the container (the hydrostatic paradox). The pressure at a given depth depends only on the fluid density, the depth, and the surface pressure — not on whether the container is wide, narrow, tilted, or U-shaped. A narrow tube and a wide tank containing the same fluid have the same pressure at the same depth.

- Misapplying hydraulic MA: confusing area ratio with force ratio. The force ratio equals the area ratio ($F_2/F_1 = A_2/A_1$), but the distance ratio is the inverse ($d_2/d_1 = A_1/A_2$). Both must be stated consistently or the energy conservation check fails.
- Using diameter instead of area. Hydraulic problems often give piston diameters. $\text{Area} = \pi r^2 = \pi(d/2)^2$. Using diameter directly instead of computing area first gives the wrong force ratio.

Common Mistake: Whenever you use $P = P_0 + \rho gh$, check: is P_0 the absolute surface pressure (P_{atm} for an open container) or a gauge pressure that must be corrected? This is the most common source of error in hydrostatic pressure problems.

Chapter Summary

Density ($\rho = m/V$) measures mass per unit volume. Pressure ($P = F/A$) measures force per unit area. Both are foundational to understanding all fluid behaviour.

- Density SI unit: kg/m^3 . Specific gravity = ρ/ρ_{water} (dimensionless; equals density in g/cm^3).
- Pressure SI unit: pascal ($\text{Pa} = \text{N/m}^2$). Acts equally in all directions at a point in a static fluid.
- Hydrostatic pressure: $P = P_0 + \rho gh$. Increases linearly with depth; independent of container shape.
- Atmospheric pressure $\approx 101\,325\text{ Pa}$ at sea level; decreases with altitude.
- Absolute = gauge + atmospheric. Gas laws and depth-pressure formulas require absolute pressure.
- Pascal's principle: pressure change in a confined fluid is transmitted undiminished throughout.
- Hydraulic systems: $F_2 = F_1(A_2/A_1)$. Force amplified by area ratio; distance reduced by same factor; work conserved.

Key Terms

fluid: Any substance that flows and deforms continuously under a shear stress; includes all liquids and gases.

density (ρ): $\rho = m/V$; mass per unit volume; SI unit kg/m^3 ; an intensive, material-

specific property.

specific gravity (SG): $SG = \rho/\rho_{\text{water}}$; dimensionless ratio of substance density to water density (1000 kg/m^3); numerically equal to density in g/cm^3 .

pressure (P): $P = F/A$; perpendicular force per unit area; SI unit pascal ($\text{Pa} = \text{N/m}^2$); acts equally in all directions at a point in a static fluid.

hydrostatic pressure: $P = P_0 + \rho gh$; the pressure at depth h in a static fluid of density ρ with surface pressure P_0 .

atmospheric pressure (P_{atm}): Pressure exerted by the weight of the atmosphere; $\approx 101\,325 \text{ Pa}$ (1 atm) at sea level; decreases with altitude.

gauge pressure: Pressure measured relative to atmospheric pressure; $P_{\text{gauge}} = P_{\text{abs}} - P_{\text{atm}}$; can be positive or negative.

absolute pressure: Pressure measured from zero (perfect vacuum); $P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}$; always non-negative.

Pascal's principle: A pressure change applied to a confined fluid is transmitted undiminished throughout the fluid and to the container walls.

hydraulic mechanical advantage: The force amplification ratio in a hydraulic system: $MA = A_{\text{out}}/A_{\text{in}} = F_{\text{out}}/F_{\text{in}}$.

hydrostatic paradox: The counterintuitive result that fluid pressure at a given depth depends only on depth, not on the shape or volume of the container.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain why a sharp knife cuts more easily than a dull knife using the concept of pressure. What changes between a sharp and a dull knife when the same force is applied?
2. Water pressure at a depth of 5 m is the same in a narrow pipe and in a wide lake. Explain why container shape does not affect pressure at a given depth.
3. Explain Pascal's principle in your own words and give one everyday example of it being used in an engineering system.
4. A hydraulic press has a force advantage of 50 (output force = $50 \times$ input force). What is the disadvantage? Why does this not violate conservation of energy?
5. Atmospheric pressure is about 101 kPa at sea level but only 26 kPa at 10 km altitude. Explain why the pressure decreases with altitude.

6. Ice has density 917 kg/m^3 and water has density 1000 kg/m^3 . What fraction of a floating iceberg is below the waterline? (Use the floating equilibrium condition from Chapter 36, but reason it out from densities here.)
 7. Distinguish absolute pressure from gauge pressure. Give an example where using gauge pressure instead of absolute pressure would lead to a serious error.
 8. Why are liquids more useful than gases for hydraulic systems? What property of liquids makes hydraulic force transmission nearly lossless?
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Density Problems

9. A solid copper sphere has radius 3.0 cm . Find its mass. ($\rho_{\text{Cu}} = 8960 \text{ kg/m}^3$; $V = (4/3)\pi r^3$)
 10. A 2.0 L bottle contains 1640 g of a liquid. Find its density in kg/m^3 and in g/cm^3 . What liquid might this be?
 11. A steel I-beam (density 7870 kg/m^3) has a cross-sectional area of 8000 mm^2 and a length of 6.0 m . Find its mass.
 12. An object with mass 450 g is submerged in water and displaces 380 cm^3 . Find its density. Does it float or sink in water?
 13. An alloy is 60% aluminium ($\rho = 2700 \text{ kg/m}^3$) and 40% copper ($\rho = 8960 \text{ kg/m}^3$) by volume. Find the density of the alloy.
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Pressure Problems

14. A concrete dam holds water to a depth of 40 m . Find the gauge pressure and absolute pressure at the base of the dam.
15. A 90 kg man stands on snowshoes with total contact area 0.12 m^2 . Find the pressure exerted on the snow. Compare to the same man standing on boots (contact area 0.024 m^2). Why do snowshoes work?
16. A submarine operates at a depth of 300 m in seawater ($\rho = 1025 \text{ kg/m}^3$). Find the absolute pressure outside the hull in Pa, kPa, and atm.
17. A mercury manometer shows a height difference of 25 cm on one side. Find the gauge pressure in Pa. ($\rho_{\text{Hg}} = 13\,600 \text{ kg/m}^3$)

18. A tyre gauge reads 220 kPa. Find the absolute pressure inside the tyre. ($P_{\text{atm}} = 101.3 \text{ kPa}$)

Fluid Pressure Exercises

19. A U-tube manometer contains water on one side and an unknown oil on the other, separated by a valve. When the valve is opened, the water side rises 3.0 cm and the oil side drops. If the oil column is 4.2 cm higher than the water column at equilibrium, find the density of the oil. (Equal pressures at the bottom: $\rho_{\text{oil}} g h_{\text{oil}} = \rho_{\text{water}} g h_{\text{water}}$; be careful with the geometry.)
20. A diver at 25 m depth in fresh water. (a) Find gauge pressure. (b) Find absolute pressure. (c) If the diver moves to 25 m depth in the ocean ($\rho = 1025 \text{ kg/m}^3$), how does the pressure change?
21. An aluminium pressure vessel (capable of withstanding 5.0 MPa gauge) contains water at 20 m depth in the ocean. What is the gauge pressure inside if the vessel is open to the seawater at 20 m depth? Is the vessel adequate?
22. Two connected tanks of different cross-sections are filled with the same liquid. Tank A has cross-sectional area 0.10 m^2 and liquid depth 0.50 m; Tank B has area 0.30 m^2 . If both tanks are at the same level and connected at the bottom, and the total volume of liquid is constant, find the liquid depth in Tank B. (Hint: pressure at the connection point must be equal.)

Hydraulic System Applications

23. A hydraulic car lift has an input piston of diameter 2.0 cm and an output piston of diameter 18 cm. (a) Find the area of each piston. (b) A force of 200 N is applied to the input. Find the output force. (c) What mass car can be lifted?
24. A hydraulic brake system has a master cylinder (bore diameter 22 mm) and four wheel cylinders (bore diameter 38 mm each). The driver applies 500 N to the brake pedal, which produces 1500 N force on the master cylinder piston. (a) Find the pressure in the brake fluid. (b) Find the force at each wheel cylinder.
25. A hydraulic press compresses a metal part. The input piston (area 4.0 cm^2) moves 120 mm. The output piston area is 600 cm^2 . (a) Find the volume of fluid transferred. (b) How far does the output piston move? (c) If 200 N is applied to

input, find the output force and the output work. Compare to the input work ($F \times d$). Is energy conserved?

26. A hydraulic system is used to lift a 2000 kg car using a pump that generates 15 MPa of pressure. What minimum output piston area is needed?

Chapter 35 Checkpoint

Before moving on to Chapter 36, confirm you can do each of the following.

I can...	See Section
Define fluid and explain how liquids differ from gases	35.1–35.2
Define density and calculate ρ , m , or V given the other two	35.3–35.4
Define and calculate specific gravity	35.5
Define pressure and calculate P , F , or A given the other two	35.6–35.7
Apply $P = P_0 + \rho gh$ to find pressure at depth in a static fluid	35.8
State the value of standard atmospheric pressure in Pa, kPa, and atm	35.9
Convert between gauge pressure and absolute pressure	35.10
State Pascal's principle and explain how it applies to confined fluids	35.11
Analyse hydraulic systems using $F_2 = F_1(A_2/A_1)$ and energy conservation	35.12
Identify and avoid common mistakes in density and pressure problems	35.13

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